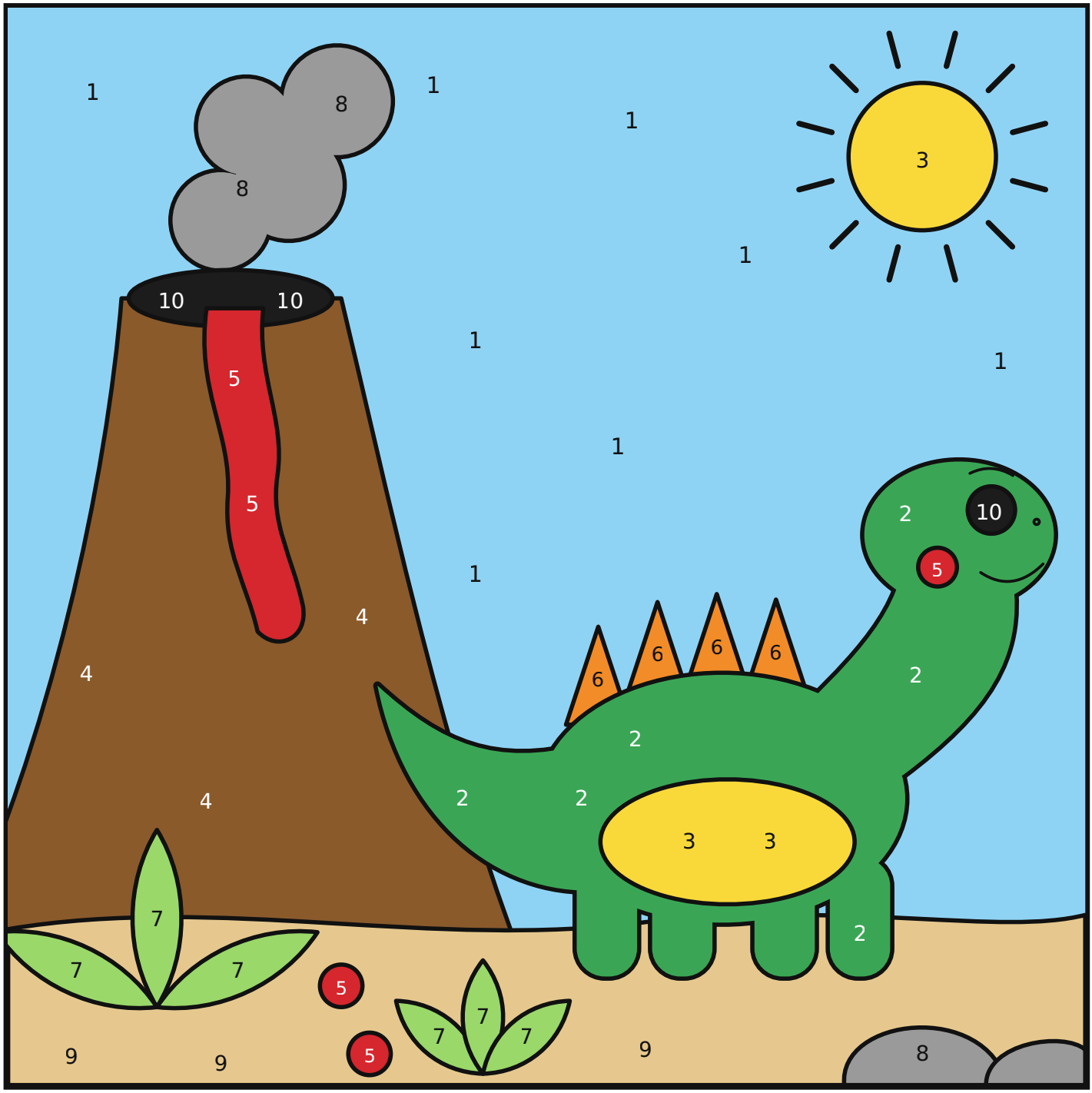
Name: **ANSWER KEY** _____ Date: _____

Systems of Equations: Color By Number



Color Code:

10 Black (1, 2)	**9** Tan (2, 1)	**6** Orange (2, −3)	**5** Red (−3, 2)	**4** Brown (3, −2)
1 Light Blue (2, 3)	**8** Gray (−3, −2)	**7** Light Green (−2, −3)	**2** Green (3, 2)	**3** Yellow (−2, 3)

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Systems of Equations: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Solve the system: $y = 2x - 1$ and $y = x + 1$

$2x - 1 = x + 1$
$x = 2$, then $y = 3$
(2, 3)

1. Color: **Light Blue**

2.
Solve the system: $y = -x + 5$ and $y = 2x - 4$

$-x + 5 = 2x - 4$
$9 = 3x$, so $x = 3$ and $y = 2$
(3, 2)

2. Color: **Green**

3.
Solve the system: $y = -x + 1$ and $y = x + 5$

$-x + 1 = x + 5$
$-4 = 2x$, so $x = -2$ and $y = 3$
(−2, 3)

3. Color: **Yellow**

4.
Solve the system: $2x + y = 4$ and $x - y = 5$

add the equations: $3x = 9$
$x = 3$, then $y = -2$
(3, −2)

4. Color: **Brown**

5.
Solve the system: $3x + 2y = -5$ and $x + y = -1$

$y = -1 - x$; $3x + 2(-1 - x) = -5$
$x - 2 = -5$, so $x = -3$ and $y = 2$
(−3, 2)

5. Color: **Red**

6.
Solve the system: $x + 2y = -4$ and $3x - y = 9$

$y = 3x - 9$; $x + 2(3x - 9) = -4$
$7x = 14$, so $x = 2$ and $y = -3$
(2, −3)

6. Color: **Orange**

7.
Solve the system: $2x - 3y = 5$ and $4x + y = -11$

multiply the second by 3 and add: $14x = -28$
$x = -2$, then $y = -3$
(−2, −3)

7. Color: **Light Green**

8.
Two numbers x and y add to -5. The first minus twice the second equals 1. Find the ordered pair (x, y).

$x + y = -5$ and $x - 2y = 1$
subtract: $3y = -6$, so $y = -2$ and $x = -3$
(−3, −2)

8. Color: **Gray**

9.
4 pens and 1 notebook cost \$9. 1 pen and 4 notebooks cost \$6. Let x = price of a pen and y = price of a notebook. Find (x, y).

$4x + y = 9$ and $x + 4y = 6$
$16x + 4y = 36$; subtract: $15x = 30$
$x = 2$ and $y = 1$
(2, 1)

9. Color: **Tan**

10.
Solve the system: $y = 3x - 1$ and $2x + y = 4$

$2x + (3x - 1) = 4$
$5x = 5$, so $x = 1$ and $y = 2$
(1, 2)

10. Color: **Black**